

Ordinary Differential Equations

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(5/10) v2.0, if you're reading this, you're trying too hard.

(6/3) **General notice:** This note only serves as a guide on how to deal with the class.

These notes are adapted from Professor C. Clark's lecture, SP25.

If you spot any error in these notes, please reach out to me via email b192.168.1.91.03@gmail.com, or you can direct message me on Discord at **3fsg**.

avoid math441 if you can

——All the victims

Methods for 1st Order Equations

Defined order and linear equations. Basic methods such as integrating factors and separation of variables. Some examples using these methods.

Proposition 1. A differential equation of the form $dy/dt = ay - b$ is solved by $y = b/a - ce^{at}$.

Example. At time $t = 0$, a tank contains 1 lb of salt dissolved in 400 L of water. Suppose that water containing 0.1 lb of salt per liter is entering the tank at a rate of 2 L/min, and that the well-stirred mixture is draining from the tank at the same rate.

Write an IVP modeling the amount of salt in lbs $Q(t)$ in the tank at time t : rate in is $0.1 \cdot 2$ and rate out is $Q/400 \cdot 2$. So $dQ/dt = 1/5 - Q/200$. The initial condition is $Q(0) = 1$.

Solution: $40 + ce^{-t/200}$. Solve for c in $1 = Q(0) = 40 + c$. $Q = 40 - 39e^{-t/200}$.

Example. Suppose an initial sum of money S_0 is deposited in a bank account with annual interest rate r . Let $S(t)$ be the value after t years. Suppose that interest compounding takes place continuously. The equation is $dS/dt = rS$. Since $S(0) = S_0$, $S = S_0 e^{rt}$.

Suppose the interest rate is 5%. How long does it take for the initial investment to be double?

We want to find the value T for which $S(T) = 2S_0$: $2S_0 = S(T) = S_0 e^{0.05T}$. Just solve for T .

Vocabulary 1.1 (order). The highest derivative in the equation.

Vocabulary 1.2 (linear differential equation). A linear n -th order ordinary differential equation is one of the form

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)}(t) + \cdots + a_1(t)y'(t) + a_0(t)y(t) = g(t)$$

where $a_n, \dots, a_0, g$ are functions of t . Otherwise, the ordinary differential equation is nonlinear.

Procedure for solving $A(t)y'(t) + B(t)y(t) = G(t)$ using the method of integrating factors

1. Rewrite in the standard form: $y'(t) + p(t)y(t) = g(t)$.
2. Compute the integrating factor: $\mu(t) = e^{\int p(t)dt}$ so that $\mu'(t) = \mu(t)p(t)$. Note that we do not add c here.
3. Multiply both sides of the standard form by $\mu(t)$: $\mu(t)y'(t) + \mu(t)p(t)y(t) = \mu(t)g(t)$.
4. Rewrite the left-hand side using the product rule: $(\mu(t)y(t))' = \mu(t)g(t)$.
5. Anti-differentiate both sides:

$$\mu(t)y(t) = \int \mu(t)g(t)dt + c$$

6. Rearrange to solve for y , and if applicable, use the initial condition to solve for c .

Example. Solve the initial value problem $(t^2 + 1)v' - tv = t$ with $v(0) = 3$.

1. Standard form:

$$v' - \frac{t}{t^2 + 1}v = \frac{t}{t^2 + 1}$$

This identifies $p(t) = \frac{t}{t^2 + 1}$.

2. The integrating factor is $\mu(t) = e^{-\int \frac{t}{t^2 + 1} dt}$. Since $\int \frac{t}{t^2 + 1} dt = \frac{1}{2} \ln(t^2 + 1)$, $\mu(t) = e^{-\frac{1}{2} \ln(t^2 + 1)} = (t^2 + 1)^{-1/2}$.
3. Multiply both sides by $\mu(t)$: $(t^2 + 1)^{-1/2}v' - t(t^2 + 1)^{-3/2}v = t(t^2 + 1)^{-3/2}$.
4. Rewrite LHS using the product rule: $((t^2 + 1)^{-1/2}v)' = t(t^2 + 1)^{-3/2}$. You know how to do this from here.

Theorem 1 (separation of variables). An ordinary differential equation of the form $dy/dx = M(x)/N(y)$ is separable because it can be written in the differential form $N(y)dy = M(x)dx$. Integrate both sides to solve this.

Existence and Uniqueness

Introduced theorems that deal with the existence and uniqueness of lower order (order less or equal to 2) equations.

Theorem 2. If $p(t)$ and $g(t)$ are continuous on an open interval (α, β) containing the point t_0 , then the initial value problem $y' + p(t)y = g(t)$, $y(t_0) = y_0$ has a unique solution on (α, β) .

Theorem 3 (existence and uniqueness for first order linear equations). If $p(t)$ and $g(t)$ are continuous on an open interval (α, β) containing the point t_0 , then the initial value problem

$$y' + p(t)y = g(t), \quad y(t_0) = y_0$$

has a unique solution on (α, β) .

Theorem 4 (existence and uniqueness for 1st order nonlinear equations). If the functions f and $\frac{\partial f}{\partial y}$ are both continuous on a rectangle $\alpha < t < \beta$, $\gamma < y < \delta$ containing (t_0, y_0) , then on some interval $(t_0 - h, t_0 + h)$ contained in (α, β) , the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0$$

has a unique solution.

Vocabulary 2.1 (bifurcation points). Consider an equation $\frac{dy}{dt} = f(a, y)$ where a is a real parameter. The critical points usually depend on a . If the number of critical points changes at an a , we say that a is a bifurcation point.

In certain special cases we can reduce the second order equation $y'' = f(t, y, y')$ to a first order equation.

Equations with y missing: For $y'' = f(t, y')$, we substitute $y' = v$ and $y'' = v'$ to get $v' = f(t, v)$.

Equations with t missing: For $y'' = f(y, y')$, we substitute $y' = \frac{dy}{dt} = v$. Interpreting v as a function of y , we get $y'' = \frac{dv}{dt} = \frac{dv}{dy} \frac{dy}{dt} = \frac{dv}{dy} v$. The new equation is $\frac{dv}{dy} v = f(y, v)$.

Theorem 5 (existence and uniqueness of linear second order equations). If the functions $p(t), q(t), g(t)$ are continuous on an open interval containing t_0 , then the initial value problem

$$y'' + p(t)y' + q(t)y = g(t), \quad y(t_0) = y_0, \quad y'(t_0) = y'_0$$

has a unique solution on the interval.

Nonhomogeneous Equations

Defined nonhomogeneous equations. Introduced theorems that describe solutions of nonhomogeneous equations. Variation of parameters and undetermined coefficients. Some examples regarding these methods.

Vocabulary 3.1 (linear nonhomogeneous of 2nd order). A linear nonhomogeneous 2nd order differential equation is of the form $L[y] = y'' + p(t)y' + q(t)y = g(t)$. To solve this, we need

- ① The complementary solution $y_c = c_1 y_1 + c_2 y_2$. This is the general solution of the homogeneous equation after setting $L[y_c] = 0$.
- ② A particular solution Y of the homogeneous equation $L[Y] = g(t)$.

Theorem 6. Suppose the characteristic equation $ar^2 + br + c = 0$ has two distinct real roots r_1 and r_2 . Then every solution of $ay'' + by' + cy = 0$ has the form $y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$.

Theorem 7. If the characteristic equation $ar^2 + br + c = 0$ has a repeated root r , then the general solution of $ay'' + by' + cy = 0$ is $y = c_1 e^{rt} + c_2 t e^{rt}$.

Theorem 8. If the characteristic equation $ar^2 + br + c = 0$ has complex roots $\alpha \pm i\beta$, then all solutions of $ay'' + by' + cy = 0$ have the form $y = e^{\alpha t}(C_1 \cos(\beta t) + C_2 \sin(\beta t))$.

Theorem 9 (nonhomogeneous superposition). The general solution of $L[y] = g(t)$ is $y = Y + y_c$.

We find the particular solution according to the following rule:

right side $g(t)$	particular solution $Y(t)$
$P_n(t)$	$t^s Q_n(t)$
$e^{\gamma t} P_n(t)$	$t^s e^{\gamma t} Q_n(t)$
$e^{\gamma t} C \cos(\beta t)$ or $e^{\gamma t} C \sin(\beta t)$	$t^s e^{\gamma t} (A \cos(\beta t) + B \sin(\beta t))$
$e^{\gamma t} P_n(t) C \cos(\beta t)$ or $e^{\gamma t} P_n(t) C \sin(\beta t)$	$t^s e^{\gamma t} (Q_n(t) \cos(\beta t) + R_n(t) \sin(\beta t))$

$s = 0, 1, 2, \dots$ is the smallest power allowing Y to avoid duplication with the complementary solution y_c . $Q_n(t)$, $P_n(t)$, $R_n(t)$ are polynomials of degree n .

Example. Solve the initial value problem $y'' - 3y' + 2y = 6t - 7$ with $y(0) = 0$, $y'(0) = 1$. We identify

y_c : The characteristic equation is $r^2 - 3r + 2 = 0$ which factors as $(r - 1)(r - 2) = 0$. So $y_c = c_1e^t + c_2e^{2t}$.

Y : Since $g(t) = 6t - 7$ is a degree 1 polynomial, we look for $Y = At + B$. This has no duplication with y_c . Now we apply the initial condition to get $Y' = A$ and $Y'' = 0$. Plugging this back into the original equation:

$$0 - 3A + 2(At + B) = 6t - 7$$

Comparing coefficients, we get $A = 3$ and $B = 1$

So the general solution is $y = 3t + 1 + c_1e^t + c_2e^{2t}$. Now solve for c_1, c_2 . $y(0) = 1 + c_1 + c_2 = 0$ and $y'(0) = 3 + c_1 + 2c_2 = 1$ give $c_1 + c_2 = -1$ and $c_1 + 2c_2 = -2$. Solving the system of equations, we find $c_1 = 0$, $c_2 = -1$. The final solution is $y = 3t + 1 - e^{2t}$.

Remark. If $g(t)$ is a sum of terms, find a particular solution for each term separately, then add them up. Note that you always need to solve for A and B . Solve c_i only when the initial condition is given.

What if our equation does not have constant coefficients, or $g(t)$ not of the form $\exp(t) \times P_n(t)^1 \times [\sin(t) \text{ or } \cos(t)]$? In that case, we use variation of parameters. Here is how we do it in steps.

I. We first find the solution $y_c = c_1y_1(t) + c_2y_2(t)$ for the homogeneous part. Then we guess the general solution to be $y = u_1(t)y_1(t) + u_2(t)y_2(t)$ where $u_1(t)$ and $u_2(t)$ are equations in t instead of constants.

II. Calculate

$$u_1'(t) = -\frac{y_2(t)g(t)}{W(y_1, y_2)(t)} \quad \text{and} \quad u_2'(t) = \frac{y_1(t)g(t)}{W(y_1, y_2)(t)}$$

where $W(y_1, y_2)(t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$. This is not 0 because $y_1(t)$ and $y_2(t)$ are linearly independent.

III. Anti-differentiate the above expressions to get $u_1(t)$ and $u_2(t)$.

Vibrations

Introduced all kinds of vibrations.

Vocabulary 4.1 (vibration). The vibration formula is $mu'' + \gamma u' + ku = F(t)$. If $\gamma = 0$, the vibration is undamped. If $F(t) = 0$, the vibration is unforced.

Remark. A vibration is unforced if and only if the corresponding differential equation is homogeneous if and only if it is *free*.

Vocabulary 4.2 (phase-amplitude form). $y = A \cos(\omega t) + B \sin(\omega t)$ can be written as $y = R \cos(\omega t - \delta)$ where the amplitude $R = \sqrt{A^2 + B^2}$, and the phase δ satisfies $\cos(\delta) = A/R$ and $\sin(\delta) = B/R$.

Vocabulary 4.3 (free undamped vibration). A free undamped vibration is given by $mu'' + ku = 0$. Rewriting the formula, we have $u'' + \omega_0^2 u = 0$ and $\omega_0 = \sqrt{\frac{k}{m}}$ is the natural frequency. The solution is $u_c(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$.

Vocabulary 4.4 (free damped vibrations). This is given by $mu'' + \gamma u' + ku = 0$. Conclude the following chart:

Case	Roots	General solution
overdamped: $\gamma > \sqrt{4km}$	distinct: $r_1, r_2 < 0$	$u = Ae^{r_1 t} + Be^{r_2 t}$
critically damped: $\gamma = \sqrt{4km}$	repeated: $-\gamma/2m < 0$	$u = (A + Bt)e^{-(\gamma/2m)t}$
under damped: $\gamma < \sqrt{4km}$	complex: $-\gamma/2m \pm i\mu$, where $\mu = \frac{\sqrt{4km - \gamma^2}}{2m}$ is the <i>quasi-frequency</i>	$u = e^{-(\gamma/2m)t} (A \cos(\mu t) + B \sin(\mu t))$ $= Re^{-(\gamma/2m)t} \cos(\mu t - \delta)$

¹Polynomials do not have negative powers.

Vocabulary 4.5 (forced undamped vibration). A forced undamped vibration is given by $mu'' + ku = F_0 \cos(\omega t)$. To solve this, we need a particular solution so that $u(t) = u_c(t) + U(t)$.

1. If $\omega_0 \neq \omega$,

$$u(t) = u_c(t) + U(t) = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t) + \frac{F_0/m}{\omega_0^2 - \omega^2} \cos(\omega t)$$

2. If $\omega_0 = \omega$,

$$u(t) = u_c(t) + U(t) = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t) + \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

The particular solution $U(t)$ in both cases is called *forced response*. If $\omega_0 \neq \omega$, $U(t)$ oscillates but it is bounded. If $\omega_0 = \omega$, $U(t)$ oscillates and it is unbounded. The amplitude of the forced response is

$$R(\omega) = \frac{F_0/m}{|\omega_0^2 - \omega^2|}$$

Vocabulary 4.6 (forced damped vibration). A forced damped vibration is given by $mu'' + \gamma u' + ku = F_0 \cos(\omega t)$. We saw that all solution of the damped ($r > 0$) free equation decay to 0. So $u_c(t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore, for large enough t , $u(t) \approx U(t)$. Its solution is given by

$$U(t) = m(\omega_0^2 - \omega^2) \frac{F_0}{\Delta^2} \cos(\omega t) + \gamma \omega \frac{F_0}{\Delta^2} \sin(\omega t)$$

where $\Delta = \sqrt{m^2(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2}$. The forced response has the phase-amplitude form $U(t) = R(\omega) \cos(\omega t - \delta)$, with amplitude F_0/Δ .

Vocabulary 4.7 (electrical vibration). The electrical vibration is given by $LQ'' + RQ' + Q/C = E(t)$. $L = m$, $R = \gamma$, and $1/C = k$ in the previous sense.

Higher Order Equations

Defined higher order equations and linear independence. Existence and uniqueness for higher order equations. Nonhomogeneous case. Series solutions.

Vocabulary 5.1 (higher order linear equations). A higher order linear equation is given by

$$L[y] = y^{(n)} + p_1(t)y^{(n-1)} + p_2(t)y^{(n-2)} + \dots + p_n(t)y = g(t)$$

with the initial condition $y(t_0) = y_0, y^{(1)}(t_0) = y_0^{(1)}, \dots, y^{(n-1)}(t_0) = y_0^{(n-1)}$.

Theorem 10 (existence and uniqueness for linear equations). If $p_1(t), \dots, p_n(t), g(t)$ are all continuous on an open interval (α, β) containing t_0 , then the initial value problem with $L[y] = g(t)$ and $y^{(i)}(t_0) = y_0^{(i)}$ has a unique solution on (α, β) .

Vocabulary 5.2 (linear independence). Functions y_i on interval I are linearly dependent if there are constants α_i (not all 0) such that $\sum_i \alpha_i y_i(t) = 0$ for some $t \in I$. Otherwise we say y_i are linear independent. On the other hand, if $W(y_1, \dots, y_n)(t) \neq 0$ for some $t \in I$, then y_i are linearly independent.

Theorem 11 (general solution of n -th order homogeneous linear equations). If $y_1, \dots, y_n$ are linearly independent solutions of $L[y] = 0$, then every solution has the form $y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$. We call such $y_1, \dots, y_n$ a *fundamental set of solutions* of $L[y] = 0$.

Example. Some examples of n -th order homogeneous linear equations:

characteristic equation	general solution
$r(r-1)(r-2) = 0$	$C_1 + C_2 e^t + C_3 e^{2t}$
$(r-1)^5 = 0$	$C_1 e^t + C_2 t e^t + C_3 t^2 e^t + C_4 t^3 e^t + C_5 t^4 e^t$
$(r^2 + 1)^3 = 0$	$e^{0t}((c_1 + c_2 t + c_3 t^2) \cos(t) + (c_4 + c_5 t + c_6 t^2) \sin(t))$
$r^3(r-2)(r^2+5)(r^2+1)^2 = 0$	$c_1 + c_2 t + c_3 t^2 + c_4 e^{2t} + c_5 \cos(\sqrt{5}t) + c_6 \sin(\sqrt{5}t) + (c_7 + c_8 t) \cos(t) + (c_9 + c_{10} t) \sin(t)$

Theorem 12 (rational roots theorem). Let $f = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be a polynomial in $\mathbb{Q}[x]$ where $a_0 \neq 0$ and $a_n \neq 0$. Then every rational root of f has the form c/d where $c|a_0$ and $d|a_n$.

Theorem 13 (nonhomogeneous equations linear equation). Suppose y_c is the general solution of $L[y] = 0$, and Y is a particular solution of $L[y] = g(t)$. Then, the general solution of $L[y] = g(t)$ is $y = y_c + Y$.

Example. For each of the following, we write out the form of the particular solution. The rule is that it should not duplicate any term of y_c .

1. $y^{(4)} - y^{(3)} = t^2$: the characteristic equation is $r^3(r-1) = 0$. So $y_c = c_1 + c_2t + c_3t^2 + c_4e^t$. $Y = t^3(At^2 + Bt + C)$.
2. $y^{(4)} + 2y^{(2)} + y = \cos(t) + t \sin(2t)$: the characteristic equation is $(r^2 + 1)^2 = 0$. So $y_c = (c_1 + c_2t) \cos(t) + (c_3 + c_4t) \sin(t)$. For $\cos(t)$, we look for $Y_1 = t^2(A \cos(t) + B \sin(t))$. And for $t \sin(2t)$, we look for $Y_2 = (Ct + D) \cos(2t) + (Et + F) \sin(2t)$. Thus $Y = Y_1 + Y_2$.

We start with $P(x)y'' + Q(x)y' + R(x)y = 0$ and $y(x_0) = y_0$, $y'(x_0) = y'_0$ where P, Q, R are polynomials, P not identically zero. The standard form is

$$y'' + \frac{Q}{P}y' + \frac{R}{P}y = 0$$

If $P(x_0) \neq 0$, we call x_0 an ordinary point. Since P, Q, R are polynomials, Q/P and R/P are continuous where $P \neq 0$, and so the initial value problem has a unique solution on an interval containing x_0 . The idea is to guess a power series solution $y = \sum_n a_n(x - x_0)^n$. We plug y into the equation and try to match the coefficients.

Example. Find the series solution of $y'' - xy' - y = 0$ with $y(0) = 0$, $y'(0) = 1$. We guess $y = \sum_{n=0}^{\infty} a_nx^n$, $y' = \sum_{n=1}^{\infty} na_nx^{n-1}$, and $y'' = \sum_{n=2}^{\infty} n(n-1)a_nx^{n-2}$. Plug all this into the equation:

$$\sum_{n=1}^{\infty} ((n+2)(n+1)a_{n+2} - na_n - a_n)x^n + 2a_2 - a_0 = 0$$

All coefficients must be 0 since the right hand side is 0. So $(n+2)(n+1)a_{n+2} - na_n - a_n = 0$ for $n \geq 1$ and $2a_2 - a_0 = 0$. The recursion formula is $a_{n+2} = (1/(n+2))a_n$ for $n \geq 0$.

Remark. We always solve for $a_{\max}$ for the recursion formula.

Systems

Studied classic models like Lotka-Volterra and SIR. Existence and uniqueness for systems. Classified systems using eigenvalues. Used matrix exponentials to solve systems.

Vocabulary 6.1 (Lotka-Volterra's predator-prey). x represents the prey population and y represents the predator population. The model is

$$\begin{aligned} x'(t) &= ax - bxy \\ y'(t) &= -cy + dxy \end{aligned}$$

The xy term corresponds to "interactions".

Vocabulary 6.2 (SIR). We assume the following:

1. Disease is of short duration and rarely fatal.
2. Disease spreads through contact between individuals.
3. Individuals who recover are immune.

Population is divided into 3 groups:

$S(t)$ = "Susceptible" = Number of people who have not been infected.

$I(t)$ = "Infected" = Number of people currently infected.

$R(t)$ = "Recovered" = Number of people who have recovered and are immune.

The total population is $S + I + R$.

The model is given by:

$$S' = -aSI \quad (1)$$

$$I' = aSI - bI \quad (2)$$

$$R' = bI \quad (3)$$

Warning: sometimes it is formulated in just (1) and (2).

Vocabulary 6.3 (equilibrium point). An equilibrium point of an autonomous system $\vec{x}'(t) = \vec{f}(\vec{x})$ is a constant vector $\vec{x}_0$ for which $\vec{f}(\vec{x}_0) = 0$. Equilibrium points are constant solutions of the system.

Vocabulary 6.4 (linear system). Linear systems look like $\vec{x}'(t) = A(t)\vec{x}(t) + \vec{g}(t)$. A is a matrix and $\vec{g}(t)$ is a vector. It is homogeneous if $\vec{g}(t) = 0$. It is autonomous if t does not show up in the actual formula. Any x, y term with degree > 1 is nonlinear. For example, x^2 and xy .

Theorem 14 (existence and uniqueness for systems). If $A(t)$ and $\vec{g}(t)$ are continuous for t on an open interval I containing t_0 , then the initial value problem

$$\vec{x}'(t) = A(t)\vec{x}(t) + \vec{g}(t), \quad \vec{x}(t_0) = \vec{x}_0$$

has a unique solution for $t \in I$. $A(t)$ and $\vec{g}(t)$ are continuous if all of their entries are continuous.

Theorem 15 (homogeneous superposition for systems). Recall that homogeneous linear systems look like $\vec{x}'(t) = A(t)\vec{x}(t)$. If $\vec{x}_1(t)$ and $\vec{x}_2(t)$ are solutions of $\vec{x}'(t) = A(t)\vec{x}(t)$ then so is $c_1\vec{x}_1(t) + c_2\vec{x}_2(t)$ for any constants c_1, c_2 .

Vocabulary 6.5 (linear independence). Vectors $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{R}^n$ are linearly dependent if there are numbers $a_1, \dots, a_k$, not all zero, such that $a_1\vec{v}_1 + \dots + a_k\vec{v}_k = \vec{0}$. Otherwise, the vectors are linearly independent. In particular, $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$ are linearly dependent if one is a real multiple of the other ($\vec{v}_1 = c\vec{v}_2$ or $\vec{v}_2 = c\vec{v}_1$). Otherwise, they are linearly independent.

Theorem 16 (span). If $\vec{v}_1, \dots, \vec{v}_n \in \mathbb{R}^n$ are linearly independent, then every vector $\vec{u} \in \mathbb{R}^n$ can be written as $\vec{u} = c_1\vec{v}_1 + \dots + c_n\vec{v}_n$ for some $c_1, \dots, c_n \in \mathbb{R}$.

Theorem 17 (general solution of homogeneous linear systems). Suppose the $n \times n$ matrix function $A(t)$ is continuous on an open interval I containing t_0 . If $\vec{x}_1, \dots, \vec{x}_n$ are solutions of $\vec{x}' = A(t)\vec{x}$ on I , and $\vec{x}_1(t_0), \dots, \vec{x}_n(t_0)$ are linearly independent, then every solution has the form $\vec{x} = c_1\vec{x}_1 + \dots + c_n\vec{x}_n$. If $\vec{v}$ is an eigenvector of A with eigenvalue λ , then the solution of $\vec{x}' = A\vec{x}$ is

$$\vec{x} = e^{\lambda t}\vec{v}$$

Vocabulary 6.6 (eigenvalues and eigenvectors). If $A\vec{v} = \lambda\vec{v}$ and $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an eigenvector with eigenvalue λ . An eigenvector is a direction that is unchanged by the transformation given by A . An eigenvalue is the scale factor of the transformation in that direction.

Theorem 18. If $\vec{u}$ and $\vec{v}$ are eigenvectors of A with different eigenvalues, then $\vec{u}$ and $\vec{v}$ are linearly independent.

Remark. Eigenvalues are given by

$$\lambda = \frac{T \pm \sqrt{T^2 - 4D}}{2}$$

where T is trace and D is determinant. Formulas for these are on the sheet.

Example (regular case). Solve $\vec{x}' = \begin{pmatrix} -4 & 6 \\ -3 & 5 \end{pmatrix} \vec{x}$. Eigenvalues are 2 and -1 .

$\boxed{\lambda = 2}$ $A - (2)I = \begin{pmatrix} -6 & 6 \\ -3 & 3 \end{pmatrix}$. We need to solve $\begin{pmatrix} -6 & 6 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. By multiplication, $\begin{pmatrix} -6p+6q \\ -3p+3q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ implies $p = q$. Pick $p = q = 1$ to get $\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$\boxed{\lambda = -1}$ $A - (-1)I = \begin{pmatrix} -3 & 6 \\ -3 & 6 \end{pmatrix}$. We need to solve $\begin{pmatrix} -3 & 6 \\ -3 & 6 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. By multiplication, $\begin{pmatrix} -3p+6q \\ -3p+6q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ implies $p = 2q$. Pick $p = 2$ and $q = 1$ to get $\vec{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

The general solution is $\vec{x} = c_1 e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Example (complex case). Solve $\vec{x}' = \begin{pmatrix} 0 & 1 \\ -2 & 1 \end{pmatrix} \vec{x}$. Eigenvalues are $1 \pm i$. We need to solve $(A - \lambda I)\vec{v} = 0$. That is $\begin{pmatrix} -1-i & 1 \\ -2 & 1-i \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. We get equations $\begin{pmatrix} (-1-i)p+q \\ -2p+(1-i)q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ that tell us $q = (1+i)p$. In this case we only need one eigenvector. So pick $p = 1$ and $\vec{v} = \begin{pmatrix} 1 \\ 1+i \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The complex solution is

$$\begin{aligned} e^{(1+i)t} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) &= e^t e^{it} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = e^t (\cos t + i \sin t) \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= e^t \cos t \begin{pmatrix} 1 \\ 1 \end{pmatrix} - e^t \sin t \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \left(e^t \cos t \begin{pmatrix} 0 \\ 1 \end{pmatrix} + e^t \sin t \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) \\ &= e^t \begin{pmatrix} \cos t \\ \cos t - \sin t \end{pmatrix} + i e^t \begin{pmatrix} \sin t \\ \cos t + \sin t \end{pmatrix} \end{aligned}$$

Since $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are independent, $c_1 e^t \begin{pmatrix} \cos t \\ \cos t - \sin t \end{pmatrix} + c_2 e^t \begin{pmatrix} \sin t \\ \cos t + \sin t \end{pmatrix}$ is the general solution.

Example (repeated case). Solve $\vec{x}' = \begin{pmatrix} 1 & 4 \\ -1 & 3 \end{pmatrix} \vec{x}$. Eigenvalues are -1 and -1 . We need to solve $\begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ which leads to the equation $\begin{pmatrix} 2p+4q \\ -p-2q \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Choose $\vec{v}_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ so that $p = -2q$. One solution is $\vec{x}_1 = e^{-t} \begin{pmatrix} -2 \\ 1 \end{pmatrix}$. To find the other solution, we need to solve

$$(A + I)\vec{v}_2 = \vec{v}_1 : \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 2p+4q \\ -p-2q \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow p = -1 - 2q$$

Pick $q = 0$. We get $\vec{v}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. So the second solution is $\vec{x}_2 = e^{-t} \left(t \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right)$. The general solution is $c_1 \vec{x}_1 + c_2 \vec{x}_2$.

Example (diagonal case). Solve $\vec{x}' = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \vec{x}$. Eigenvalues are 5 and 5 . Since every vector is an eigenvector, we pick $\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. These are linearly independent. So the general solution is $c_1 e^{5t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{5t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Theorem 19 (phase plane uniqueness theorem). If two phase plane trajectories of $\vec{x}' = A\vec{x}$ intersect at some point, the trajectories follow exactly the same curve in the plane. Simply put: phase plane trajectories cannot cross.

Now we classify some phase portrait of real distinct eigenvalues $\lambda_1 \neq \lambda_2$.

$\lambda_1 < 0 < \lambda_2$ this is a saddle (unstable) with the general solution $c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$. The first term dominates as $t \rightarrow -\infty$ and the second term dominates as $t \rightarrow \infty$.

$0 < \lambda_1 < \lambda_2$ this is a nodal source (unstable) with the general solution $c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$. The first term dominates as $t \rightarrow -\infty$ and the second term dominates as $t \rightarrow \infty$.

$\lambda_1 < \lambda_2 < 0$ this is a nodal sink (stable) with the general solution $c_1 e^{\lambda_1 t} \vec{v}_1 + c_2 e^{\lambda_2 t} \vec{v}_2$. The first term dominates as $t \rightarrow -\infty$ and the second term dominates as $t \rightarrow \infty$.

Remark. To draw these, first draw all the eigenvectors. Then draw the "U" shaped curves according to the dominating term in the solution as $t \rightarrow \infty$.

Now we classify some phase portrait of complex² eigenvalues $a \pm ib$.

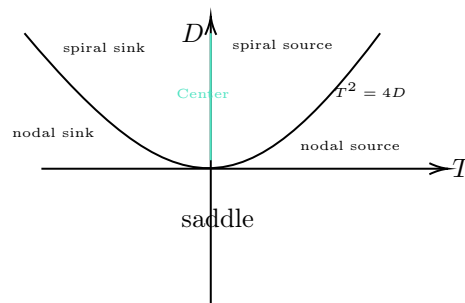
$a = 0$ is a center. Trajectories are ellipses centered at the origin.

$a > 0$ is a spiral source. The solution is same as for $a = 0$, but scaled by a factor of e^{at} . So the phase plots spiral away from the origin.

$a < 0$ is a spiral sink. The solution is same as for $a = 0$, but scaled by a factor of e^{at} . So the phase plots spiral toward the origin.

In terms of directions, check the direction of $A\vec{x}$ at $\vec{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. If it is pointing upward, then we are going counter clockwise. If it is pointing downward, then we are going clockwise.

Summary of phase portrait classification: trace-determinant plane



Vocabulary 6.7 (matrix exponential). Let A be a square matrix.

$$e^A := I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{A^n}{n!}$$

Example. Let $A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$. $e^A = \sum_{n=0}^{\infty} \frac{1}{n!} \begin{pmatrix} a^n & 0 \\ 0 & b^n \end{pmatrix} = \begin{pmatrix} e^a & 0 \\ 0 & e^b \end{pmatrix}$.

Proposition 2 (law of exponents for commuting matrices). Suppose A and B are square matrices of the same size. If A and B commute, then $e^{A+B} = e^A e^B$.

²I understand that all real numbers are also complex. But you know what I am talking about. Please shut the fuck up.

Proposition 3 (inverse of matrix exp). $(e^A)^{-1} = e^{-A}$.

Proposition 4 (derivative of matrix exp). Let t be some real number. $\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$.

Theorem 20 (constant coefficient linear systems using matrix exp). Suppose we have a system $\vec{x}' = A\vec{x}$ with initial vector $\vec{x}(0) = \vec{x}_0$. $\vec{x}(t) = e^{At}\vec{x}_0$ solves the system.

Theorem 21 (nonhomogeneous constant coefficient linear system using matrix exp). Let A be an invertible matrix. The system $\vec{x}' = A\vec{x} - \vec{b}$ is solved by $\vec{x} = A^{-1}\vec{b} + e^{At}\vec{c}$ where $\vec{c}$ is a constant that comes from integrating both sides of $(e^{-At}\vec{x})' = -e^{-At}\vec{b}$.

Linearization

Studied how to linearize systems. And some examples.

We analyze nonlinear autonomous system near its equilibrium points by "linearizing" it. The original differential equation and the linearized differential equation have the same phase line near 0. The linearized differential equation is simpler. So it is easier to analyze. **1-dimensional with equilibrium at $x_0 = 0$**

nonlinear	linearized	method
$x' = x - x^2$	$x' = x$	kicked
$x' = -2x + x^3 - 4x^5$	$x' = -2x$	kicked
$x' = 3\sin x$	$x' = 3x$	$\sin x = x - \frac{x^3}{3!} + \dots$
$x' = e^{7x} - 1$	$x' = 7x$	$e^{7x} = 1 + 7x + \frac{7x^2}{2!} + \dots$

1-dimensional with equilibrium at $0 \neq x_0 \in \mathbb{R}$

Replace $f(x)$ with its linear approximation (tangent line) at x_0 : the linearized equation is $x' = f'(x_0)(x - x_0) + f(x_0) = f'(x_0)(x - x_0)$. $f(x_0) = 0$.

nonlinear	x_0	linearized	shifted $u = x - x_0$
① $x' = x^2 - 4$	2	$x' = 4(x - 2)$	$u' = 4u$
② $x' = \sin x - 1/2$	$\pi/6$	$x' = \cos(\pi/6)(x - \pi/6)$	$u' = \cos(\pi/6)u$
③ $x' = e^{2x} - e^2$	1	$x' = 2e^2(x - 1)$	$u' = 2e^2u$

Change of variables to shift equilibrium to 0: $u = x - x_0$. So $(u' = x')$.

- ① $f(x) = x^2 - 4 \Rightarrow f'(x) = 2x \Rightarrow f'(2) = 4$.
- ② $f(x) = \sin x - \frac{1}{2} \Rightarrow f'(x) = \cos x \Rightarrow f'(\pi/6) = \cos(\pi/6)$
- ③ $f(x) = e^{2x} - e^2 \Rightarrow f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$

2-dimensional with equilibrium at (x_0, y_0)

Theorem 22. Replace f and g with their tangent plane approximations, then shift the variables with $u = x - x_0$ and $v = y - y_0$ to move the equilibrium to the origin. The linearized and shifted system is:

$$\begin{cases} u' = \frac{\partial f}{\partial x}(x_0, y_0)u + \frac{\partial f}{\partial y}(x_0, y_0)v \\ v' = \frac{\partial g}{\partial x}(x_0, y_0)u + \frac{\partial g}{\partial y}(x_0, y_0)v \end{cases}$$

or $\vec{u}' = J\vec{u}$, where $J = \begin{pmatrix} \frac{\partial f}{\partial x}(x_0, y_0) & \frac{\partial f}{\partial y}(x_0, y_0) \\ \frac{\partial g}{\partial x}(x_0, y_0) & \frac{\partial g}{\partial y}(x_0, y_0) \end{pmatrix}$ is the Jacobian matrix at (x_0, y_0) , and $\vec{u} = \begin{pmatrix} u \\ v \end{pmatrix}$.

Vocabulary 7.1 (generic phase portrait). "Generic" phase portraits are saddle, nodal sink, nodal source, spiral sink, and spiral source. All others are "non-generic". They are the "borderline" cases in the T-D plane.

Theorem 23 (linearization theorem). If the linearized system has a generic phase portrait, then the nonlinear system has a qualitatively similar phase portrait near the equilibrium point.

Proposition 5 (polar stuff). We investigate the nonlinear system in polar coordinates r and ϑ through the change of variable $x = r \cos \vartheta$ and $y = r \sin \vartheta$. Then naturally $\vartheta = \arctan(y/x)$ and $r^2 = x^2 + y^2$. In particular,

$$\begin{aligned}\vartheta' &= \frac{1}{1 + (y/x)^2} \cdot \left(\frac{xy' - x'y}{x^2} \right) \\ 2rr' &= 2xx' + 2yy'\end{aligned}$$

This gives us three cases

- ① When $r' > 0$. Trajectories spiral outward.
- ② When $r' < 0$. Trajectories spiral inward.
- ③ When $r' = 0$. The trajectory is a circle of radius r .

Example. Consider the system $\begin{cases} x' = (y + x/5)(1 - x^2) = f \\ y' = -x(1 - y^2) = g \end{cases}$

- (a) Find all equilibrium points.
 - (b) Linearize at each equilibrium point, and classify the phase portrait.
 - (c) Try to sketch the phase portrait of the original system.
- (a): First we set $f = 0$: $y = -x/5$, $x = 1$, $x = -1$. Then we set $g = 0$: $x = 0$, $y = 1$, $y = -1$. They correspond to points $(0, 0)$, $(-5, 1)$, $(1, 1)$, $(-1, 1)$, $(5, -1)$, $(1, -1)$, $(-1, -1)$.
- (b): Given $f = (y + x/5)(1 - x^2)$ and $g = -x(1 - y^2)$, the Jacobian matrix is:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{1}{5}(1 - x^2) + (y + \frac{x}{5})(-2x) & 1 - x^2 \\ -(1 - y^2) & 2xy \end{pmatrix}$$

To get the system at each point, plug in each of those 7 points above for x and y . Then classify each system.

(c): It turned out all of them have generic phase portraits. We know the behavior near the equilibria of the original system will be qualitatively similar.

Vocabulary 7.2 (invariant set). A set S in the phase plane is (positively) invariant if

$$\vec{x}(t_0) \in S \Rightarrow \vec{x}(t) \in S, \quad \forall t \geq t_0$$

In other words, if the solution starts in or enters S , then it remains in S . A horizontal line $y = a$ is invariant if $y' = 0$ on that line. A vertical line $x = a$ is invariant if $x' = 0$ on that line.

Vocabulary 7.3 (nullcline). We use nullcline to analyze nonlinear autonomous systems

x -nullcline: A curve in the phase plane where $x' = 0$. Direction field vectors on an x -nullcline point straight up or down.

y -nullcline: A curve in the phase plane where $y' = 0$. Direction field vectors point only left or right.

Points at which an x -nullcline intersects a y -nullcline are the equilibrium points.

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